

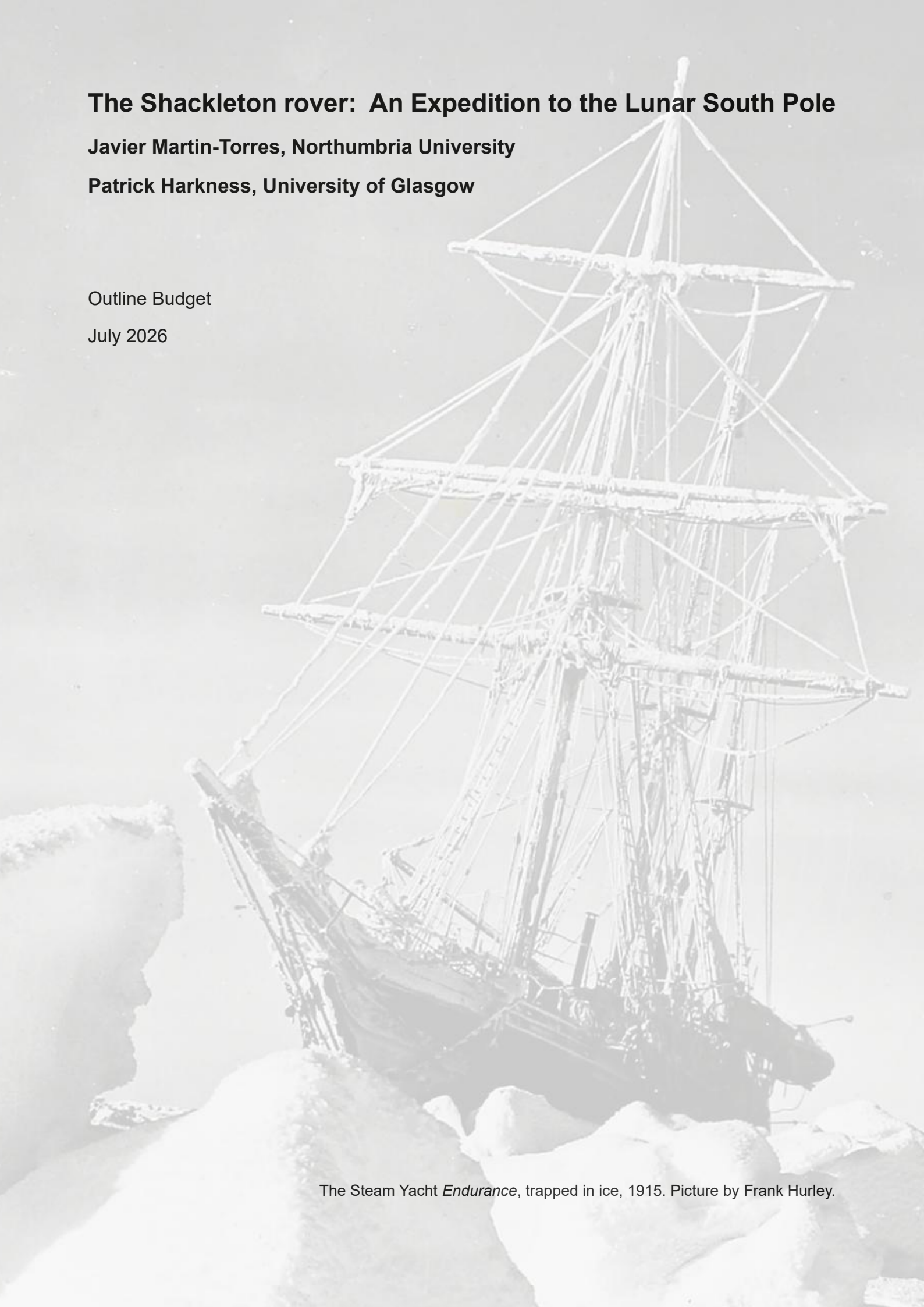
The Shackleton rover: An Expedition to the Lunar South Pole

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Outline Budget

July 2026



The Steam Yacht *Endurance*, trapped in ice, 1915. Picture by Frank Hurley.

Costs: We estimate that the hardware can built and operated for two weeks for a total investment of around £22m. A CLPS delivery to the Moon is likely on the order of £3m.

COST SUMMARY BY CATEGORY			
Category	Cost (£)	% of Total	Notes / Heritage
Rover Platform (Structure, Mobility, Power, Thermal, Comms, Avionics)	£6,500,000	29.5%	Northumbria, Leicester, Open Univ?
Scientific Payloads (Mass Spectrometer Suite + Sonicated Penetrator)	£7,000,000	31.8%	Open Univ (MS heritage from Peregrine); Glasgow/Leeds/BAS (penetrator)
Assembly, Integration, Test & Verification (AITV)	£2,200,000	10.0%	Glasgow dirty-vacuum, Leeds simulant tests, full environmental qual
Project Management, Systems Engineering & Assurance	£1,800,000	8.2%	Lead: University of Glasgow / consortium coordination
Ground Segment, Software, Launch Integration & Ops Preparation	£2,700,000	12.3%	Goonhilly, international partner ops ??? (Canada/Japan/Australia)
Mission Operations, Data Analysis & Science Support	£1,000,000	4.5%	14-day 24/7 ops with partner rotation; quick-look science
Education, Training, Public Engagement & Outreach	£500,000	2.3%	PhD cohort aligned to 4-year programme; STEM inspiration
Contingency & Risk Reserve	£300,000	1.4%	~1.4% reserve; higher TRL risk on novel penetrator mitigated by testing
TOTAL ESTIMATED MISSION COST	£22,000,000	100.0%	Benchmark: Australian Roo-ver A\$42M (~£22M) for equivalent scope

WBS	Category / Element	Cost (£)	%	Primary UK Lead / Heritage & Notes
1.0	PROJECT MANAGEMENT, SYSTEMS ENGINEERING & ASSURANCE	£1,800,000		University of Glasgow (lead); consortium of all signatories
1.1	Project Management & Control	£600,000		Full lifecycle PM, risk registers, schedule/cost control
1.2	Systems Engineering & Interface Control	£700,000		Requirements flow-down, ICDs with CLPS provider, verification planning
1.3	Quality Assurance, Safety & Reviews	£500,000		PDR/CDR/TRR/FRR; planetary protection compliance
2.0	ROVER PLATFORM	£6,500,000		Northumbria, Leicester, Open?
2.1	Structure & Chassis (0.4m x 0.15m sides)	£1,200,000		Lightweight composite/metal; sized for 0.25m ² side solar panels
2.2	Mobility System (4 wheels, skid-steer, gearmotors)	£1,800,000		~750g per corner; low-mass wheels for lunar regolith
2.3	Power System (Solar + Battery)	£1,100,000		0.25m ² panels (~40W), 50Wh Li-ion; 3kg/m ² & 200Wh/kg assumptions
2.4	Thermal Control (Insulation + PCM + RHU)	£700,000		Phase-change material for cold-soak sorties to shadowed regions
2.5	Communications (S-band, deployable mast, patch antenna)	£900,000		Directional patch on yaw-controlled mast (Astromast/Rolatable heritage)
2.6	Avionics, Onboard Computer & Harness	£800,000		Nominal 2.5% mass allocation; radiation-tolerant COTS where possible
3.0	SCIENTIFIC PAYLOADS	£7,000,000		Core science return
3.1	Mass Spectrometer Suite (TG-EGA)	£4,200,000		Open University ??; adaptation of Exospheric MS / PITMS (Peregrine heritage). 4kg. Stepped heating, ion trap, evolved gas analysis for H ₂ O discrimination (adsorbed / pore / cement) + co-volatiles
3.2	Sonicated Penetrator System	£2,800,000		Glasgow?? (ultrasonics lead), Leeds?? (granular modelling), BAS ?? (icy drilling). 3kg to 20cm depth. Standing-wave fluidisation reduces penetration force by ~10x in granular media. TRL ~4.5; thermal-vac testing critical. Includes heaters, thermistors, ultrasonic transducer, penetration sensors
4.0	ASSEMBLY, INTEGRATION, TEST & VERIFICATION	£2,200,000		UK national facilities
4.1	Integration & Functional Testing	£800,000		Cleanroom assembly, functional checkout, software-in-the-loop
4.2	Environmental Testing (Vibe, TVAC, EMI/EMC)	£700,000		Standard qual for lunar environment
4.3	Regolith & Penetrator Performance Testing	£700,000		Glasgow dirty-vacuum chamber (one of Europe's largest); Leeds granular flow models; icy-regolith analogues; calibration of force reduction & volatile release
5.0	GROUND SEGMENT, SOFTWARE & LAUNCH INTEGRATION	£2,700,000		
5.1	Flight Software & Autonomy	£900,000		Path planning from mast imagery, obstacle avoidance, autonomous sortie execution
5.2	Ground Control Software & UK Ground Station	£700,000		Goonhilly dedicated command capability; full-spectrum ops demo
5.3	CLPS Integration, ICDs, Mission Assurance	£700,000		Payload accommodation, mechanical/electrical interfaces, safety reviews
5.4	International Partner Coordination	£400,000		Ops hand-over with Canadian/Japanese/Australian teams for 24h coverage
6.0	MISSION OPERATIONS & DATA ANALYSIS	£1,000,000		
6.1	Surface Operations (14 days, 24/7)	£600,000		Real-time commanding, health monitoring, sortie planning; partner-supported shifts
6.2	Data Processing, Archiving & Science Quick-Look	£400,000		Volatile profiles, thermal properties, regolith strength; public data release
7.0	EDUCATION, TRAINING & OUTREACH	£500,000		Critical for skills pipeline
7.1	PhD Studentships & Early-Career Training	£350,000		Aligned to 4-year timeline; hands-on hardware + ops experience
7.2	Public Engagement & STEM Inspiration	£150,000		National narrative (Shackleton spirit); school/university outreach; media
8.0	CONTINGENCY & RISK RESERVE	£300,000		~1.4% of total; applied to highest-risk elements (penetrator TRL advancement)
	GRAND TOTAL	£22,000,000	100%	

The profile follows a typical project curve.

Category	Year 1 (Concept/Feasibility)	Year 2 (Prelim & Detailed Design)	Year 3 (Build, IATV Peak)	Year 4 (Final Test, Launch, Ops)	TOTAL
Project Management, SE & Assurance	£450,000	£600,000	£500,000	£250,000	£1,800,000
Rover Platform	£800,000	£2,000,000	£2,800,000	£900,000	£6,500,000
Scientific Payloads (MS + Penetrator)	£1,200,000	£2,500,000	£2,500,000	£800,000	£7,000,000
Assembly, Integration, Test & Verification	£200,000	£400,000	£1,200,000	£400,000	£2,200,000
Ground Segment, Software & Launch Integration	£300,000	£700,000	£900,000	£800,000	£2,700,000
Mission Operations & Data Analysis	£0	£0	£200,000	£800,000	£1,000,000
Education, Training & Outreach	£150,000	£150,000	£100,000	£100,000	£500,000
Contingency & Risk Reserve	£50,000	£100,000	£100,000	£50,000	£300,000
ANNUAL TOTAL	£3,150,000	£6,450,000	£8,300,000	£4,100,000	£22,000,000
% of Total Mission Cost	14.3%	29.3%	37.7%	18.6%	100%